1. Two identical solid spheres each of mass $2\ \mathrm{kg}$ and radii $10\ \mathrm{cm}$ are fixed at the ends of a light rod. The separation between the centres of the spheres is $40\ \mathrm{cm}$. The moment of inertia of the system about an axis perpendicular to the rod passing through its middle point is______ $\times 10^{-3}\ \mathrm{kg\ m}^2$.
[2023 (06 Apr Shift 1)]

2. A ring and a solid sphere rotating about an axis passing through their centres have same radii of gyration. The axis of rotation is perpendicular to plane of ring. The ratio of radius of ring to that of sphere is $\sqrt{\frac{2}{x}}$. The value of x is _____.
[2023 (06 Apr Shift 2)]

3. The moment of inertia of a semicircular ring about an axis, passing through the center and perpendicular to the plane of ring, is $\frac{1}{\mathrm{x}}\ \mathrm{MR}^2$, where R is the radius and M is the mass of the semicircular ring. The value of x will be _______.
[2023 (08 Apr Shift 1)]

4. A force of $-P\hat{k}$ acts on the origin of the coordinate system. The torque about the point $(2,-3)$ is $P\left(a\hat{i}+b\hat{j}\right)$, The ratio of $\frac{a}{b}$ is $\frac{x}{2}$. The value of x is
[2023 (10 Apr Shift 2)]

5. A solid sphere of mass 500 g radius $5\ \mathrm{cm}$ is rotated about one of its diameter with angular speed of $10\ \mathrm{rad\ s}^{-1}$. If the moment of inertia of the sphere about its tangent is $x\times 10^{-2}$ times its angular momentum about the diameter. Then the value of x will be
[2023 (11 Apr Shift 1)]

6. A circular plate is rotating in horizontal plane, about an axis passing through its centre and perpendicular to the plate, with an angular velocity ω. A person sits at the centre having two dumbbells in his hands. When he stretched out his hands, the moment of inertia of the system becomes triple. If E be the initial Kinetic energy of the system, then final Kinetic energy will be $\frac{E}{x}$. The value of x is
[2023 (11 Apr Shift 2)]

7. For rolling spherical shell, the ratio of rotational kinetic energy and total kinetic energy is $\frac{x}{5}$. The value of x is _____.
[2023 (12 Apr Shift 1)]

8. A solid sphere is rolling on a horizontal plane without slipping. If the ratio of angular momentum about axis of rotation of the sphere to the total energy of moving sphere is $\pi:22$ then, the value of its angular speed will be_________rad s^{-1}.
[2023 (13 Apr Shift 1)]

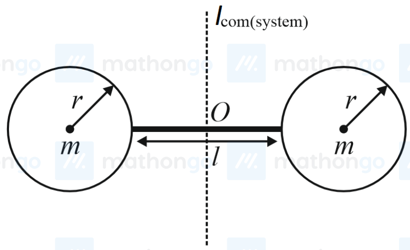
9. A light rope is wound around a hollow cylinder of mass $5\ \mathrm{kg}$ and radius $70\ \mathrm{cm}$. The rope is pulled with a force of 52.5 N. The angular acceleration of the cylinder will be _____ $\mathrm{rad\ s}^{-2}$.
[2023 (13 Apr Shift 2)]

10. A solid sphere and a solid cylinder of same mass and radius are rolling on a horizontal surface without slipping. The ratio of their radius of gyrations respectively $\left(k_{sph}:k_{cyl}\right)$ is $2:\sqrt{x}$. The value of x is _______.
[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (176)	**2.** (5)	**3.** (1)	**4.** (3)	**5.** (35)	**6.** (3)	**7.** (2)	**8.** (4)
9. (15)	**10.** (5)						

1. (176)



The moment of inertia of a sphere is given by the formula $I = \frac{2}{5}MR^2$.

Since there are two masses, the total moment of inertia about the axis perpendicular to the rod passing through the centre is

$I = 2 \times \left[\frac{2}{5}MR^2 + M\left(\frac{l}{2} + r\right)^2\right] \ldots (i)$

The given data is

$M = 2\ \text{kg}$

$l = (0.4 - 0.2)\ \text{m} = 0.2\ \text{m}$

$\text{R} = 0.1\ \text{m}$

Substituting the values in equation (i)

$I = \left(\frac{4}{5}\left(2 \times 0.1^2\right) + 2 \times 2\left(\frac{0.2}{2} + 0.1\right)^2\right)\ \text{kg m}^2$

$\Rightarrow I = \left(\left(\frac{4}{5} \times 0.02\right) + (4 \times 0.04)\right)\ \text{kg m}^2$

$\Rightarrow I = (0.016 + 0.16)\ \text{kg m}^2 = 0.176\ \text{kg m}^2$

2. (5)

The moment of inertia of the ring about an axis perpendicular to the plane of ring is $I_r = mk^2$.

The moment of inertia of the sphere is

$I_s = \frac{2}{5}mR^2{}_s$

It is given that

$mk^2 = mR^2{}_r = \frac{2}{5}mR_s{}^2 \ldots (i)$

From equation (i)

$R_r = \sqrt{\frac{2}{5}}R_s$

From the question, it can be written that

$\frac{R_r}{R_s} = \sqrt{\frac{2}{x}}$

$\Rightarrow \frac{R_r}{R_s} = \sqrt{\frac{2}{5}}$

$\Rightarrow x = 5$

3. (1)

The moment of inertia of a semicircular ring about a line perpendicular to the plane of the ring and passing through its centre is given as I= MR^2, where m and R are the mass and radius of the ring. Comparing the given value

.I = $\text{MR}^2 = \frac{1}{x}MR^2$

$\Rightarrow \text{x} = 1$

4. (3)

It is given that the force $\vec{F} = -P\hat{k}$
The coordinates of the origin is $(0, 0)$.
The coordinates of the point given is $(2, -3)$
The displacement vector is
$\vec{r} = \left(0-2\right)\hat{i} + \left(0+3\right)\hat{j} = -2\hat{i} + 3\hat{j}$
Hence, the torque is
$\vec{\tau} = \vec{r} \times \vec{F} = \left(-2\hat{i} + 3\hat{j}\right) \times \left(-P\hat{k}\right)$
$\Rightarrow \vec{\tau} = -2P\hat{j} - 3P\hat{i} = P\left(a\hat{i} + b\hat{j}\right)$
So, it can be written
a $= -3$, $b = -2$
$\therefore\ \frac{a}{b} = \left(\frac{3}{2}\right) \Rightarrow$ so $x = 3$

5. (35)

The angular momentum about the diameter is
$L_{\text{diameter}} = \frac{2}{5}MR^2\omega;$
The moment of inertia of a sphere is $I_{CM} = \frac{2}{5}MR^2$. The moment of inertia about the tangent is
$I_{\text{tangent}} = I_{CM} + MR^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$.
It is given that $\omega = 10 \text{ rad s}^{-1}$
Using the given relation between the moment of inertia and the angular momentum,
$\frac{7}{5}MR^2 = x \times 10^{-2} \times \frac{2}{5}MR^2\omega$
$\Rightarrow x = \frac{3.5 \times 10^2}{\omega} = 3.5 \times 10 = 35$

6. (3)

The kinetic energy is given by
$E = \frac{1}{2}I\omega^2 = \left(\frac{L^2}{2I}\right)$
It is given that the new moment of inertia
$I' = 3I$
Clearly, the final kinetic energy is
$E_f = \frac{L^2}{2I'} = \frac{L^2}{6I}$
$\Rightarrow E_f = \left(\frac{E}{3}\right)$

7. (2)

The rotational kinetic energy is given by
$K_{rot} = \frac{I\omega^2}{2} = \frac{1}{3}MR^2\omega^2.$
The total kinetic energy is given by
$K_{tot} = \frac{1}{2}mv^2 + \frac{1}{3}MR^2\omega^2$
$= \frac{mR^2\omega^2}{2} + \frac{1}{3}MR^2\omega^2$
$= \frac{5}{6}MR^2\omega^2$
Thus, the ratio is
$\frac{K_{rot}}{K_{tot}} = \frac{\frac{1}{3}MR^2\omega^2}{\frac{5}{6}MR^2\omega^2} = \frac{2}{5}$
Comparing with the given value,
$\frac{x}{5} = \frac{2}{5}$
$\Rightarrow x = 2$

8. (4)

The angular momentum is given by

$L = I_{CM}\omega = \frac{2MR^2\omega}{5}$

The total kinetic energy is

$E = \frac{1}{2}\left(Mv^2 + \frac{2MR^2\omega^2}{5}\right)$

The ratio of angular momentum and kinetic energy is

$\frac{L}{E} = \frac{I_{CM}\omega}{\frac{1}{2}Mv^2+\frac{1}{2}\cdot\frac{2}{5}MR^2\omega^2} = \frac{\frac{2}{5}MR^2\omega}{\frac{7}{10}MR^2\omega^2}$

$\Rightarrow \frac{\pi}{22} = \frac{4}{7}\cdot\frac{1}{\omega} \Rightarrow \omega = 4 \text{ rad s}^{-1}$

9. (15)

The formula to calculate the moment of inertia (I) of the cylinder about its radial axis is given by

$I = Mr^2 \quad \ldots(1)$

Also, the torque (τ) on the cylinder about the central axis because of the application of the external force is given by

$\tau = Fr \quad \ldots(2)$

Also, the torque can be expressed as

$\tau = I\alpha \quad \ldots(3)$

Substitute the expressions from equation (1) and (2) into equation (3) and simplify to obtain the angular acceleration.

$Fr = Mr^2\alpha$

$\Rightarrow \alpha = \frac{F}{Mr} \quad \ldots(4)$

Substitute the values of the known parameters into equation (4) to calculate the required angular acceleration.

$\alpha = \frac{52.5 \text{ N}}{5 \text{ kg} \times 0.70 \text{ m}}$

$= 15 \text{ rad s}^{-2}$

10. (5)

Considering rotational axis as the diametrical axis for sphere and axis of cylinder. Then

$K_1^2 = \frac{2}{5}R^2$ and $K_2^2 = \frac{1}{2}R^2$

$\therefore \frac{K_1}{K_2} = \sqrt{\frac{2/5}{1/2}} = \sqrt{\frac{4}{5}}$

$\Rightarrow \frac{K_1}{K_2} = \frac{2}{\sqrt{5}}$

Comparing it with the given value in the question,

$\frac{2}{\sqrt{5}} = \frac{2}{\sqrt{x}}$

$\therefore x = 5$